

Name: AVIRAJ KUMAR RAI AVIRAJ KUMAR RAI | Reg. No: RA2511003011026

Unit 4 : Session 9 : SLO 2 | TCP and UDP Socket Types

I. Match the Following

1. TCP — Reliable, ordered: connection-oriented byte stream with acknowledgements, retransmission and sequencing.
2. UDP — Fast, no guarantee: connectionless datagrams with no delivery or ordering guarantee, minimal overhead.
3. Port — Logical communication endpoint: a 16-bit number identifying a specific process/service on a host.

II. Key Notes

- TCP use cases: web (HTTP/HTTPS), email, file transfer — where correctness matters.
- UDP use cases: live video/voice, DNS, online gaming — where speed matters more than perfection.
- A socket = IP address + port + protocol; TCP sockets are SOCK_STREAM, UDP sockets are SOCK_DGRAM.